Geometric series

1 Find the sum of the first 5 terms of the series: $24 - 12 + 6 \ldots$
Leave your answer correct to the nearest whole number.

2 Find the sum of the first 7 terms of the geometric series: $64 + 16 + 4 \ldots$

3 Find the sum of the first 7 terms of the series: $8 + 1 + \frac{1}{8} \ldots$
Round your answer to three decimal places.

4 Find the sum of the first 11 terms of the series: $1 - 2 + 4 \ldots$

5 Find the sum of the first 12 terms of the series: $5 + 10 + 20 \ldots$

6 Find the sum of the first 5 terms of the geometric sequence defined by the following. Round your answers to two decimal places.

a $u_1 = 2.187$ and $r = 1.134$

b $u_1 = -4.186$ and $r = -2.848$

7 For each of the following series:

i Find n, the number of terms in the series.

ii Find the sum of the series.

a $4 - 8 + 16 - \ldots - 2048$

b $64 + 16 + 4 + \ldots + \frac{1}{16}$

8 Find n, the number of terms, in each of the following series:

a The sum of n terms in the geometric series $2 + 10 + 50 + \ldots$ is $195\,312$.

b The sum of n terms in the geometric series $5, -20, 80, \ldots$ is $262\,145$.

c The sum of n terms in the geometric series $16 + 4 + 1 + \ldots$ is $\frac{1365}{64}$.

d The sum of n terms in the geometric series $40, -8, \frac{8}{5}, \ldots$ is $\frac{20\,832}{625}$.

9 Consider the series: $5 + \frac{5}{2} + \frac{5}{4} \ldots$

a Find the common ratio, r.

b Find the sum of the first 10 terms, rounding your answer to one decimal place.

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10 The first 3 terms of a sequence are $x,\ \frac{2}{5}x^2,\ \frac{4}{25}x^3$.

a Write a simplified expression for the sum of the first n terms of this sequence.

b Evaluate the sum when $x = 5$ and $n = 9$.

11 Consider the series: $\frac{1}{4} - \frac{1}{5} + \frac{1}{16} - \frac{1}{25} + \frac{1}{64} - \frac{1}{125} + \ldots$
Form an expression for the sum of the first $2n$ terms of the series.

12 The sum of the first 6 terms of a geometric series is 28 times the sum of its first 3 terms.
Find r, the common ratio.

13 The sum of the first 9 terms of a geometric series is $436\,905$, and the common ratio is $\frac{1}{4}$.
Find u_1, the first term in the series.

Summation notation

14 Write the following series using summation notation:

a $1 + 2 + 4 + \ldots + 512$

b $1 - \frac{1}{2} + \frac{1}{4} - \frac{1}{8} + \ldots + \frac{1}{1024}$

c $-3 + 9 - 27 + \ldots + 729$

d $6 + 18 + 54 + \ldots + 486$

e $\frac{1}{16} + \frac{1}{32} + \frac{1}{64} + \ldots + \frac{1}{1024}$

f $8 + 4 + 2 + \ldots + \frac{1}{8}$

15 Evaluate:

a $\sum_{i=1}^{4} 4^i$

b $\sum_{j=1}^{5} 243\left(\frac{5}{3}\right)^j$

c $\sum_{j=1}^{5} 32\left(\frac{1}{2}\right)^j$

d $\sum_{i=3}^{8} (-3)^i$

e $\sum_{i=2}^{7} \left(-2^i\right)$

f $\sum_{i=1}^{5} \left(2\,(3)^{i-1}\right)$

g $\sum_{j=4}^{8} \left(\frac{1}{4}\,(3)^{j-1}\right)$

h $\sum_{j=1}^{4} \left(-3\left(\frac{1}{3}\right)^j\right)$

16 Write the following series using summation notation:

$$x + 2x^4 + 4x^7 + 8x^{10} + \ldots 2^{18}x^{55}$$

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Technology

17 Use the sequence facility of your calculator to determine for which term number n the geometric sequence $u_n = 6\left(2.1^{n-1}\right)$ has a sum of 217.3146.

18 Use the sequence facility of your calculator to determine for which term number n the geometric sequence $u_n = 800\left(0.9^{n-1}\right)$ has a sum of $4556.262\,32$.

19 The 5th term of a geometric sequence is 405. The sum of the 6th, 7th and 8th terms is $15\,795$.

- **a** Write an equation involving u_1, the first term, and r, the common ratio, of this geometric sequence for the 5th term.
- **b** Write an equation involving u_1, the first term, and r, the common ratio, and the sum of the 6th, 7th and 8th terms of the sequence.
- **c** Use the simultaneous solving facility of your calculator to find the values of u_1 and r. Assume u_1 and r are positive.
- **d** Hence, state the general term, u_n, of the sequence.

Applications

20 Suppose you save $\$1$ the first day of a month, $\$2$ the second day, $\$4$ the third day, $\$8$ the fourth day, and so on. That is, each day you save twice as much as you did the day before.

- **a** How much will your total savings be for the first 13 days?
- **b** How much will your total savings be for the first 29 days?

21 Average annual salaries are expected to increase by 5% each year. The average annual salary this year is found to be $\$49\,000$.

- **a** Calculate the expected average annual salary in 4 years.
- **b** This year, Aaron starts at a new job in which he will receive the average annual salary for each year of his employment. Over the coming 4 years (including this year), he plans to save half of each year's annual salary.
  Determine his total savings over these 4 years.

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22 This year, 600 people are expected to enter the workforce as registered nurses. This number is expected to increase by 4% next year, and increase by the same percentage every year after that.
Calculate the following, rounding your answers to the nearest whole number:

a The number of nurses expected to enter the workforce between six and seven years from now.

b The number of nurses expected to enter the workforce over the next six years.

23 A conveyor belt is being used to remove materials from a quarry. Every thirty minutes, the conveyor belt empties out $\frac{1}{5}$ of whatever material remains in the quarry.
The quarry initially holds $14\,000\ \text{m}^3$ of materials.

a How much material has been taken out after 90 minutes?

b How much material is left in the quarry after 90 minutes?

24 The first blow of a hammer drives a post a distance of $64\ \text{cm}$ into the ground. Each successive blow drives the post $\frac{3}{4}$ as far as the preceding blow. In order for the post to become stable, it needs to be driven $\frac{781}{4}\ \text{cm}$ into the ground.
If n is the number of hammer strikes needed for the pole to become stable, find n.

25 A car's brakes failed and the driver immediately turned the engine of the car off. In the first second after the engine was shut off, the car travelled $20\ \text{m}$. Every successive second, the car travelled 80% of the distance covered in the previous second.

a Find an expression for the total distance the car travelled in the first n seconds

b Hence, find the total distance the car travelled in the first four seconds to two decimal places.

c Traffic had built up $109\ \text{m}$ away from where the driver turned off the engine.
How far from the traffic build-up did the car come to rest?

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26 At the start of 2014, Pauline deposits $\$5000$ into an investment account. At the end of each quarter, she makes an extra deposit of $\$700$.
The table below shows the first few quarters of 2014. All values in the table are in dollars:

Quarter	Opening Balance	Interest	Deposit	Closing Balance
Jan - Mar	5000	200	700	5900
Apr - Jun	5900	236.00	700	6836.00
Jul - Sep	6836.00	273.44	700	7809.44

a Find the quarterly interest rate.

b Write an expression for the amount in the account at the end of the first quarter.

c Hence, write an expression for the amount in the account at the end of the second quarter.

d Hence, write a similar expression for the amount in the account at the end of the third quarter.

e Write an expression for the amount in the investment account after n quarters.

f Hence, determine the total amount in Pauline's account at the beginning of 2016 to the nearest dollar.

27 Iain opens an account to help save for a house. He opens the account at the beginning of 2017 with an initial deposit of $\$50\,000$ that is compounded annually at a rate of 5.9% per annum. He makes further deposits of $\$3000$ at the end of each year.

a Write an expression for the amount in the account after:

i 1 year **ii** 2 years **iii** 3 years

b The amount in the account after n years can be expressed as:

$$50\,000 \times (1.059)^n + 3000 \times (1.059)^{n-1} + \ldots + 3000 \times (1.059)^2 + 3000 \times 1.059 + 3000$$

This can be written as $50\,000 \times (1.059)^n$, plus the sum of a geometric sequence. Write an expression representing this sum.

c Hence, determine the total value of Iain's savings in 2024, to the nearest dollar.

28 Lisa invests $\$30\,000$ at a rate of 1.5% per month compounded monthly. Each month, she withdraws $\$600$ from her investment after the interest is paid and the balance is reinvested in the account.

a Write an expression for A_1, the balance of the account after 1 month.

b Write an expression for:

i A_2 **ii** A_3

c Hence, write an expression for A_n.

d Calculate the amount Lisa has saved after 3 years.

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29 To save up to buy a car, Laura opens a savings account that earns 6% per annum compounded monthly.
She initially deposits $\$1400$ when she opens the account at the beginning of the month, and then deposits $\$165$ at the end of every month.

a The amount in the account after n months can be expressed as the nth term of a geometric sequence plus the sum of a different geometric sequence.
Write an expression for the amount in the investment account after n months.

b Hence, determine the amount Laura has saved after 3 years.

30 Rochelle invests $\$190\,000$ at a rate of 7% per annum compounded annually, and wants to work out how much she can withdraw each year to ensure the investment lasts 20 years.

a Write an expression for A_1, the balance of the account after 1 year. Use x to represent the amount to be withdrawn each year.

b Write an expression for:

i A_2 **ii** A_3

c Hence, write an expression for A_n.

d Hence, determine Rochelle's annual withdrawal amount.

31 Mario invests $\$110\,000$ at a rate of 16% per annum compounded quarterly and wants the investment to last 25 years.

a The amount in the account after n quarters can be expressed as the nth term of a geometric sequence minus the sum of a different geometric sequence.
Write an expression for the amount in the investment account after n quarters. Use x to represent the amount to be withdrawn each quarter.

b Hence, determine Mario's quarterly withdrawal amount.

32 Emma invests $\$190\,000$ at a rate of 3% per annum compounded monthly, and wants to work out how much she can withdraw each month to ensure the investment lasts 30 years.

a The amount in the account after n months can be expressed as the nth term of a geometric sequence minus the sum of a different geometric sequence.
Write an expression for the amount in the investment account after n months. Use x to represent the amount to be withdrawn each month.

b Hence, determine Emma's monthly withdrawal amount.

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